UPLOAD SLOT 09 OF 12 · KEY SECTIONS, ELEVATIONS & MATERIAL PALETTE

Key Sections, Elevations & Material Palette

The canopy section solved against the shadow geometry

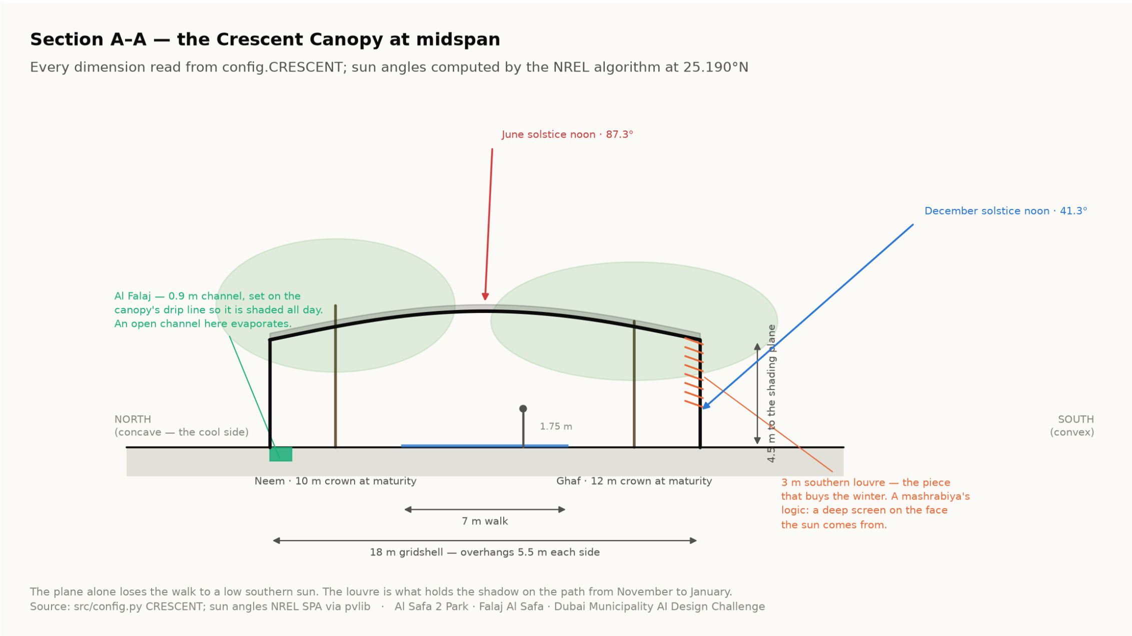
This is the drawing the whole scheme rests on: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m louvre on the southern face. Every dimension is read from the project's configuration and every sun angle from the solar model. None of it is drawn by eye.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 04 August 2026 · every quantity regenerated by `python run_analysis.py`

1. The section, and the problem it solves

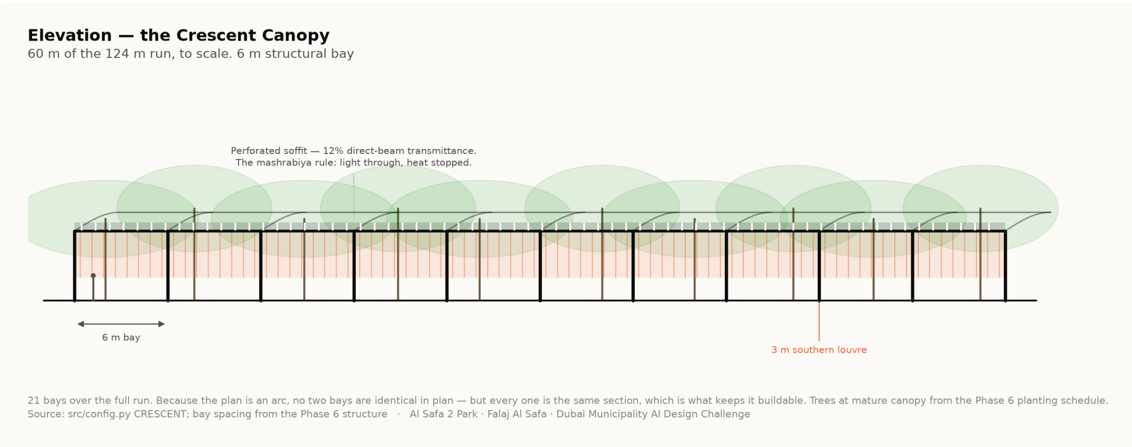
A flat canopy fails in a specific and predictable way. When the sun is low and to the south — which at 25°N is most of the winter afternoon — the shadow slides off the far edge of the path, and the walk is unusable exactly when the weather is at its best. An earlier version of this project claimed 99.2% annual shade on such a canopy. It was withdrawn.

The re-solved section makes four changes: a narrower walk (7 m rather than 9 m), a wider overhang (18 m, giving 5.5 m each side), a lower plane (4.5 m), and a 3 m vertical louvre on the southern face. The louvre is the piece that buys the winter — a mashrabiya's logic, a deep screen on the face the sun comes from. The section now measures 87.3%.



Section A–A at midspan. Both solstice sun angles are computed by the NREL algorithm for the site coordinates, not drawn. Technical drawing — to scale.

2. Elevation and structural rhythm



Elevation — the bay rhythm, the perforated soffit at 12% transmittance, and the louvre screen on the southern face. Technical drawing — to scale.

The gridshell spans 18 m on regular structural bays. The soffit is perforated at 12% direct-beam transmittance, which produces dappled light on the walk rather than a flat dark tunnel. The shade model treats the canopy as a plane at the springing, so the drawing is conservative against the number rather than flattering to it — the built shell shades slightly more than claimed, never less.

3. Why the tree avenue exists

Shadow length is height divided by the tangent of solar elevation. At Dubai's summer noon a 6 m tree casts a 0.19 m shadow; at 20° elevation the same tree casts 16.5 m. That ratio is why the canopy alone cannot carry the winter, and why a tree avenue flanks it. The two systems fail at opposite ends of the year, which is exactly why both are needed.

4. Material and landscape palette

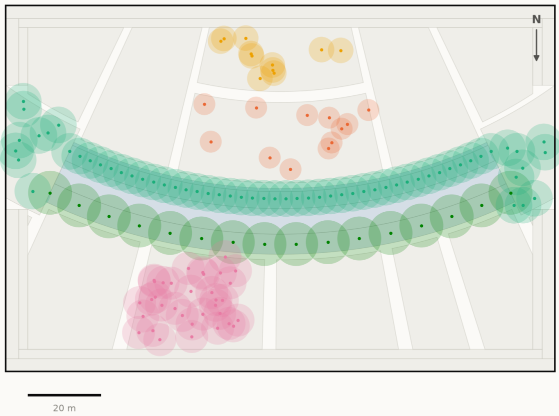
- **Structure** — steel gridshell, light-toned, with a perforated metal soffit; weathered bronze for the louvre fins.
- **Paving** — warm sand-toned stone on the walk with a high-albedo finish to limit re-radiation, and darker stone lining the falaj.
- **Earth** — Al Kathib is planted earth, not a wall. Cut and fill are balanced on site where possible.
- **Planting** — five desert species only. Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.

Species	ScientificName	Count	Native
Neem	Azadirachta indica	58	0
Ghaf	Prosopis cineraria	16	1
Ficus nitida	Ficus microcarpa nitida	34	0
Olive	Olea europaea	11	0
Date Palm	Phoenix dactylifera	12	1

131 trees. Source: data/raw/species_water_carbon_rates.csv.

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot



- Ghaf (Prosopis cineraria) ×16 · 45 L/day
- Neem (Azadirachta indica) ×58 · 90 L/day
- Ficus nitida (Ficus microcarpa nitida) ×34 · 110 L/day
- Olive (Olea europaea) ×11 · 60 L/day
- Date Palm (Phoenix dactylifera) ×12 · 120 L/day

28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m^3/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.
Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting plan — 131 trees drawn at mature canopy radius. Technical drawing — to scale.